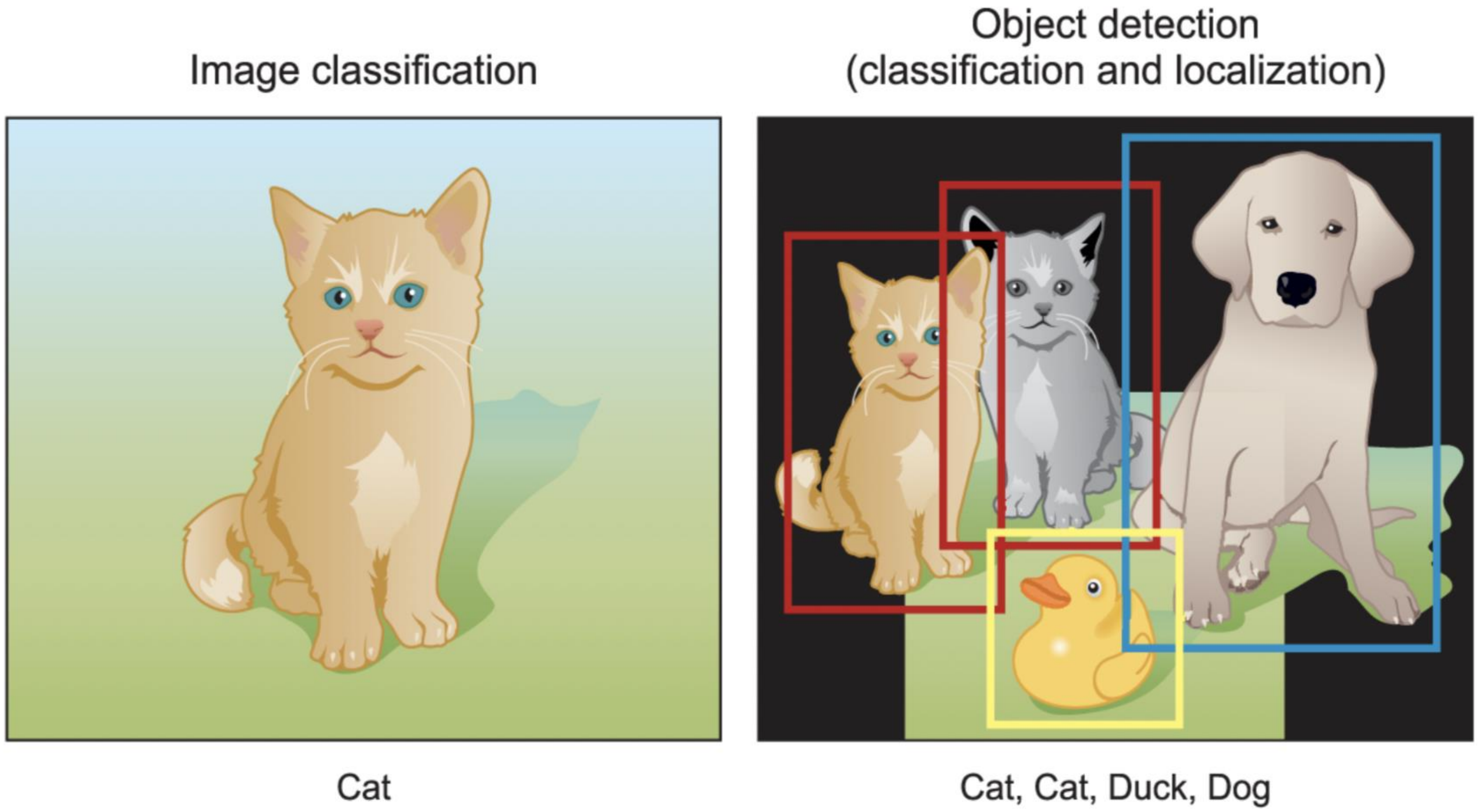
Lecture 14

Object detection with R-CNN, SSD, and YOLO

Classification Vs Object Detection



Classification Vs Object Detection

Image classification	Object detection

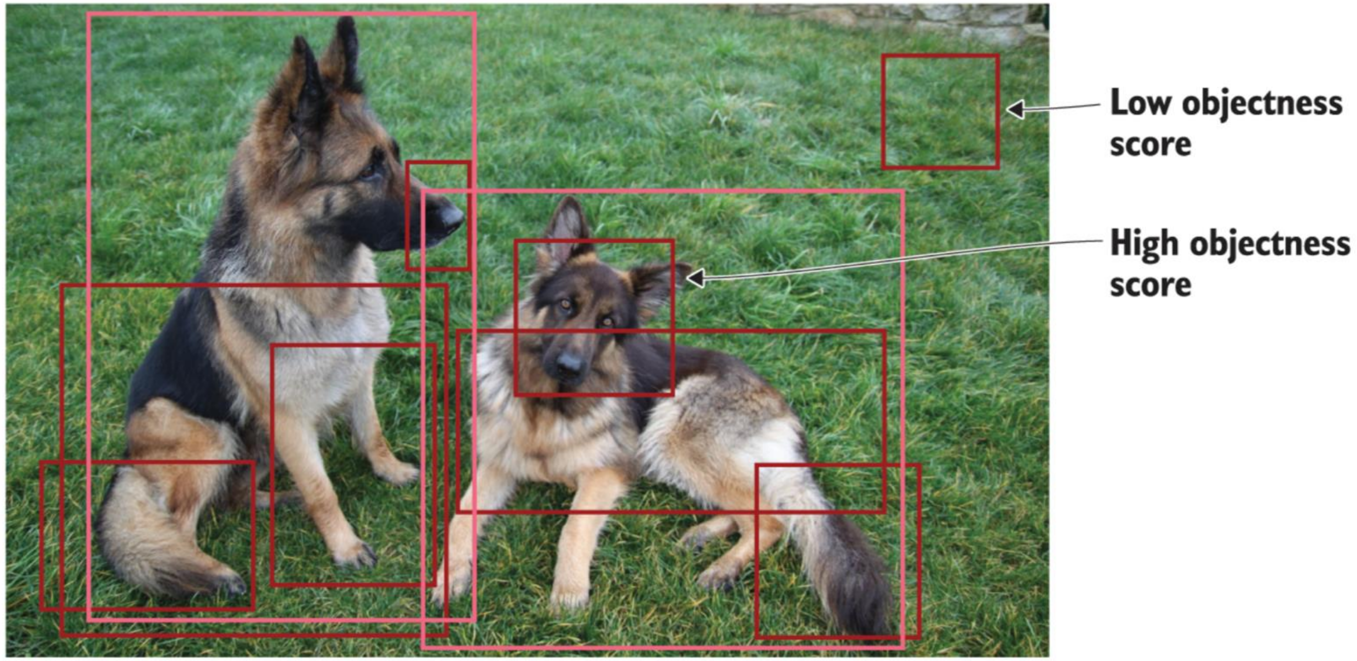
The goal is to predict the type or class of an object in an image.
 	
- Input: an image with a single object
- Output: a class label (cat, dog, etc.)
- Example output: class probability (for example, 84% cat)
	The goal is to predict the location of objects in an image via bounding boxes and the classes of the located objects.
	
	- Input: an image with one or more objects
	- Output: one or more bounding boxes (defined by coordinates) and a class label for each bounding box
	- Example output for an image with two objects:
	  – box1 coordinates (x, y, w, h) and class probability
	  – box2 coordinates and class probability
	
	Note that the image coordinates (x, y, w, h) are as follows: (x and y) are the coordinates of the bounding-box center point, and (w and h) are the width and height of the box.

General object detection framework Components

- Region Proposal
- Feature extraction and network predictions
- Non-maximum suppression (NMS)
- Evaluation metrics

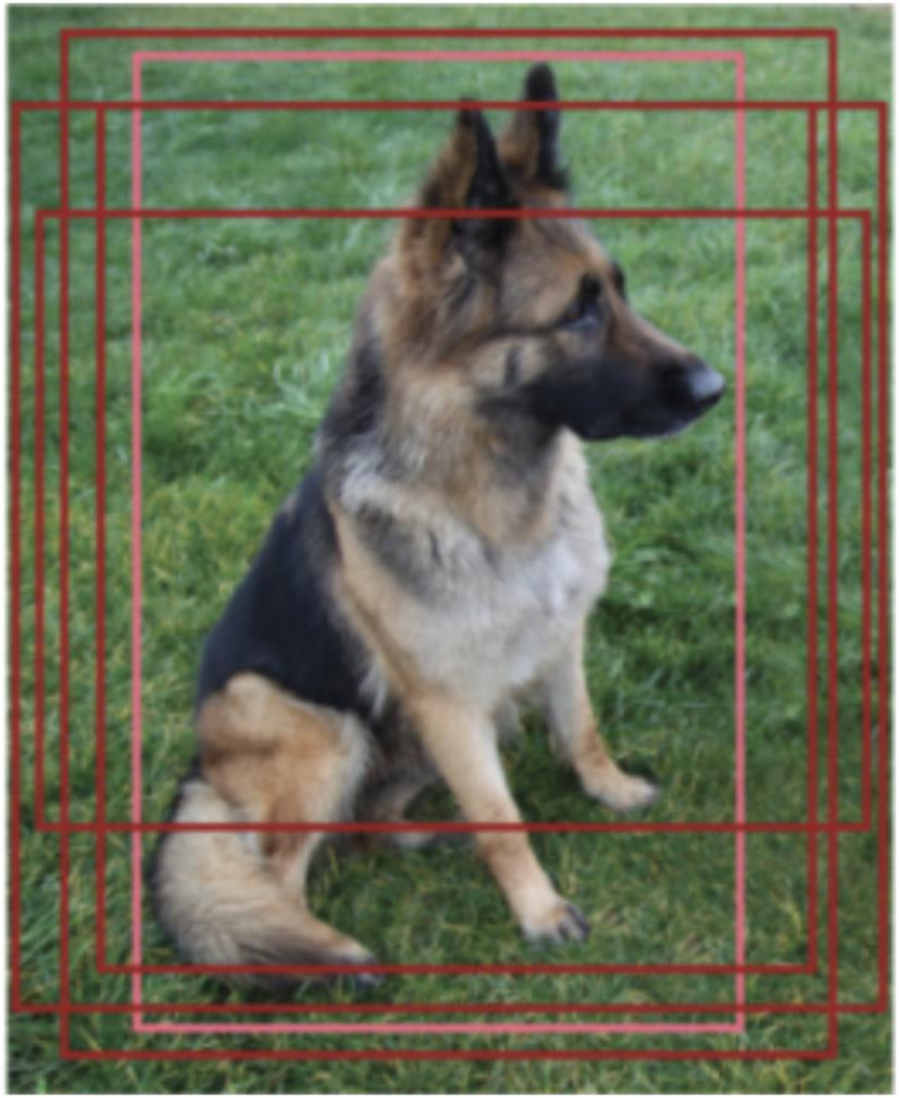
Region Proposals

- Assigns objectness score
- Region proposal generation
  - Selective search algorithm
  - Complex visual feature extraction using deep neural networks
- Produces a number of bounding boxes for analysis
  - regions are accepted/rejected based on the objectness score
- Computationally expensive



Network Predictions

- Uses image classification model to extract visual features
- Predictions
  - Bounding box
    - A tuple {x,y,w,h}
      - {x,y} – bounding box center point coordinates
      - w – box width
      - h – box height
  - Class corresponding to the object in the image
- Can predict multiple bounding boxes for the same object if multiple RoIs are predicted from the region proposal step



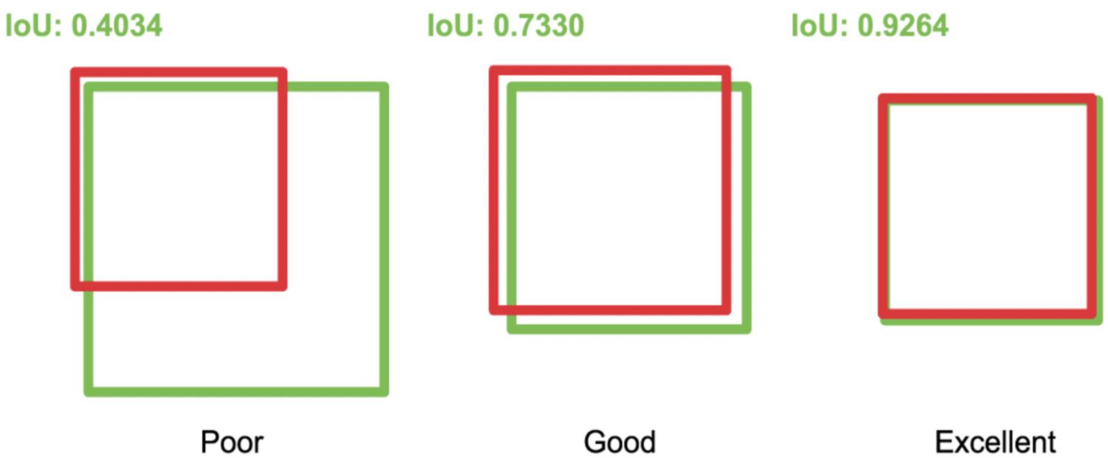
Non-maximum suppression (NMS)

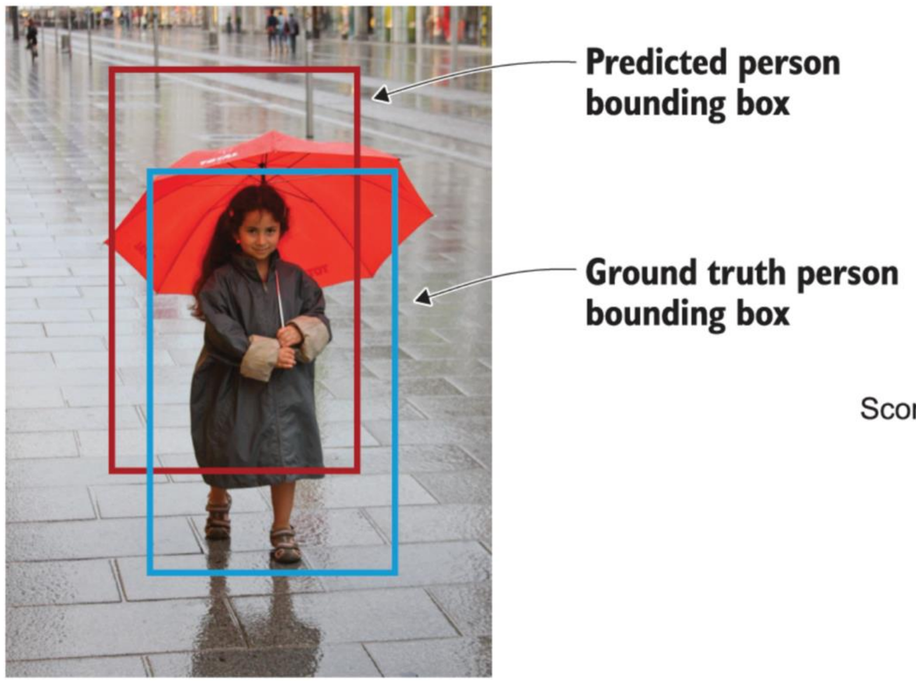
- Finds a single bounding box for the object in the image
- Checks the object's maximum prediction probability from the surrounding boxes
  - Eliminate the bounding boxes comparing the predictions with the confidence threshold
  - Select the bounding box with the highest probability from remaining boxes
  - Check for the overlap between remaining boxes with same class predictions
    - Use Intersection over Union (IoU) to average out the bounding boxes with high overlap
  - Eliminate the boxes with IoU value less than NMS threshold
- Hyperparameters to tune: confidence threshold and NMS threshold

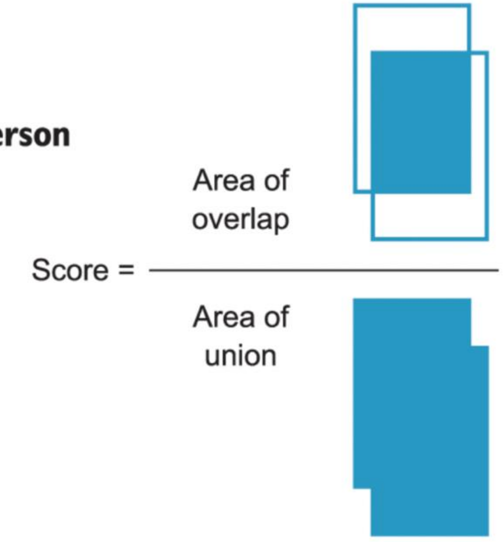
Object Detection Evaluation Metrics

- FPS: frame per second
  - Measures object detection speed
- mAP: Mean Average Precision
  - Better object detection for higher values
- Intersection over Union (IoU)
  - Measure the overlap between 2 bounding boxes

Intersection Over Union (IoU)





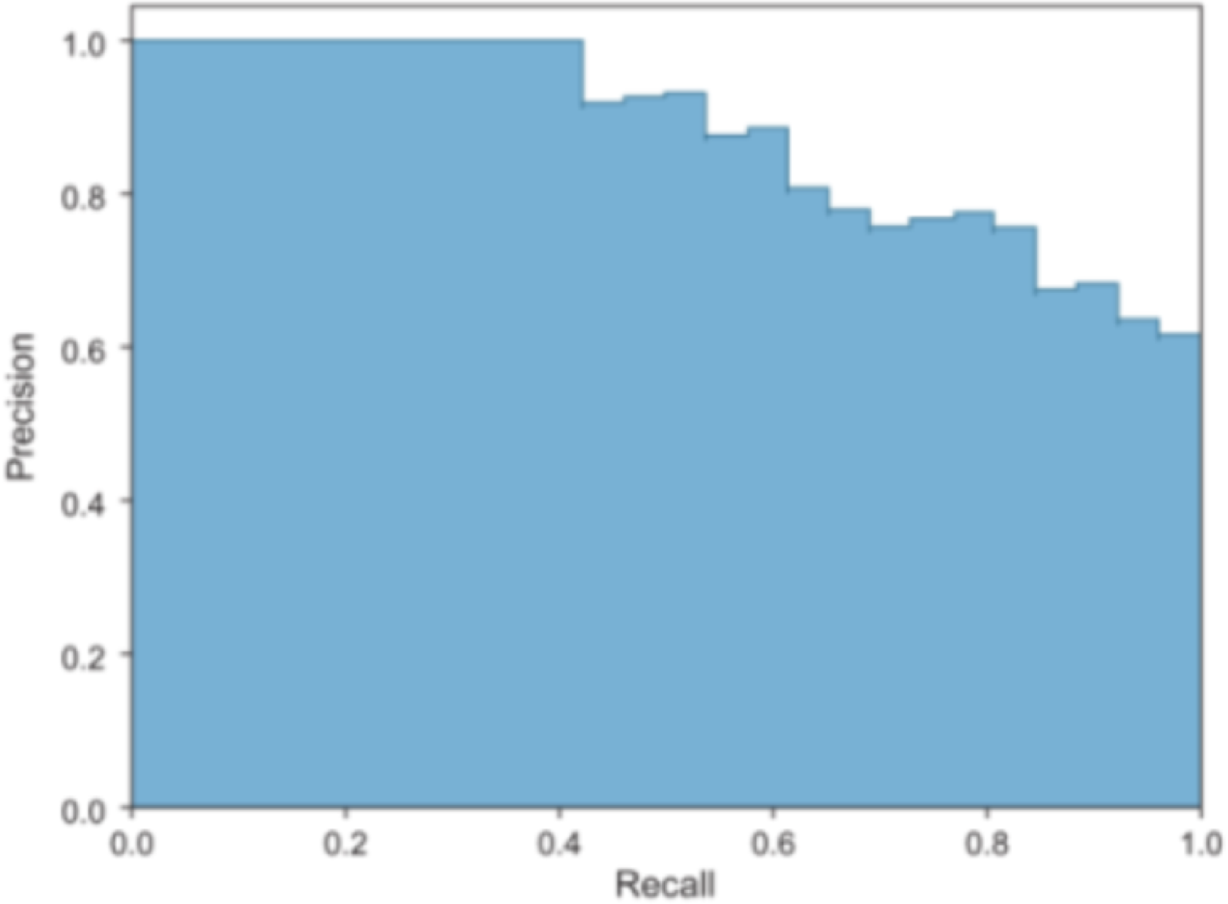


- $IoU = \dfrac{B_{groundtruth} \cap B_{predicted}}{B_{groundtruth} \cup B_{predicted}}$

Precision-recall curve (PR curve)

- $Recall = \frac{TP}{TP+FN}$ = Sensitivity
- $Precision = \frac{TP}{TP+FP}$ = Specificity

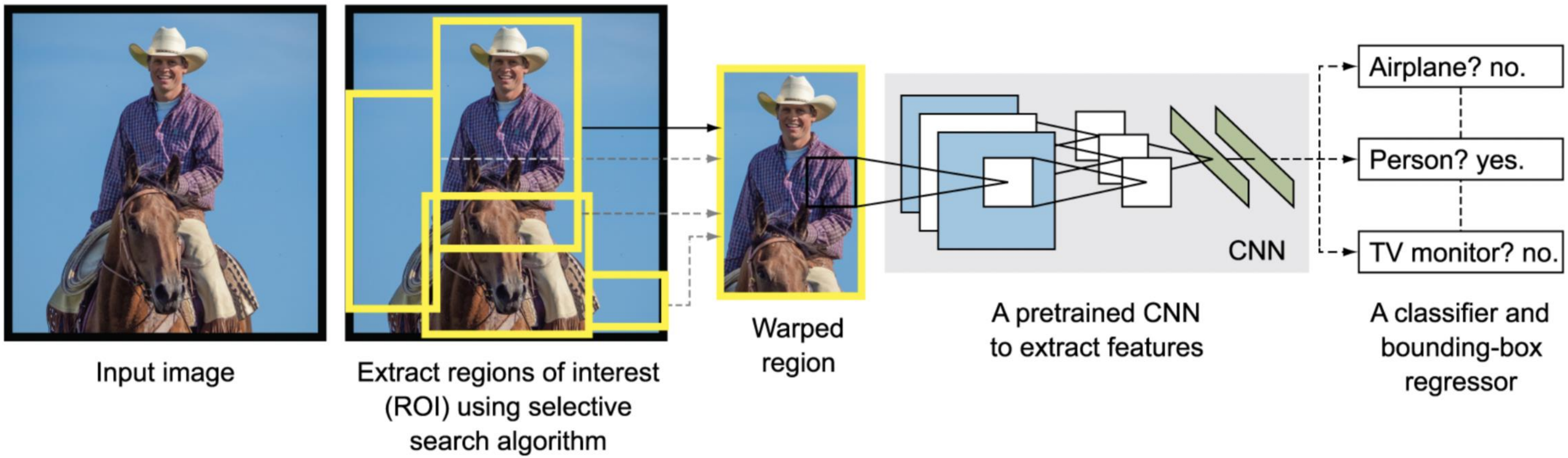
- Precision and Recall values for a detector should stay high irrespective of change in the confidence threshold.



mAP (Mean Average Precision)

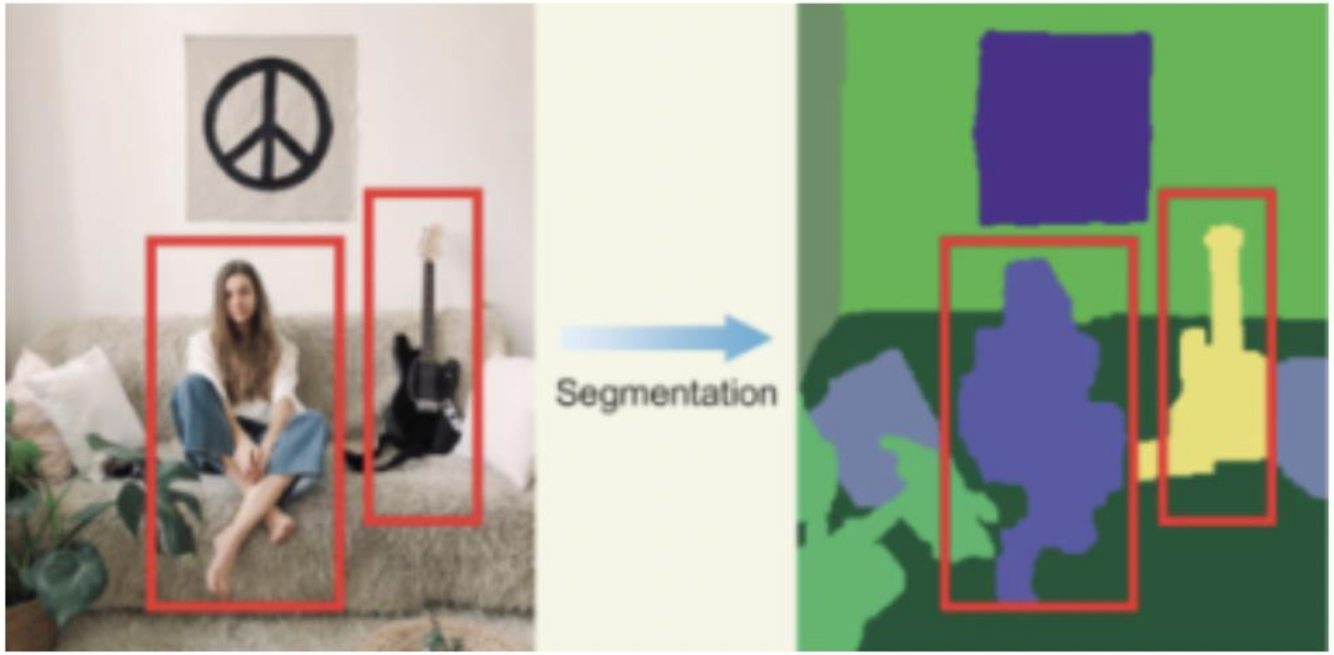
- Each bounding box will have an objectness score associated (probability of the box containing an object).
- Precision and recall are calculated.
- Precision-recall curve (PR curve) is computed for each class by varying the score threshold.
- Calculate the average precision (AP): it is the area under the PR curve. In this step, the AP is computed for each class.
- Calculate mAP: the average AP over all the different classes.

R-CNN

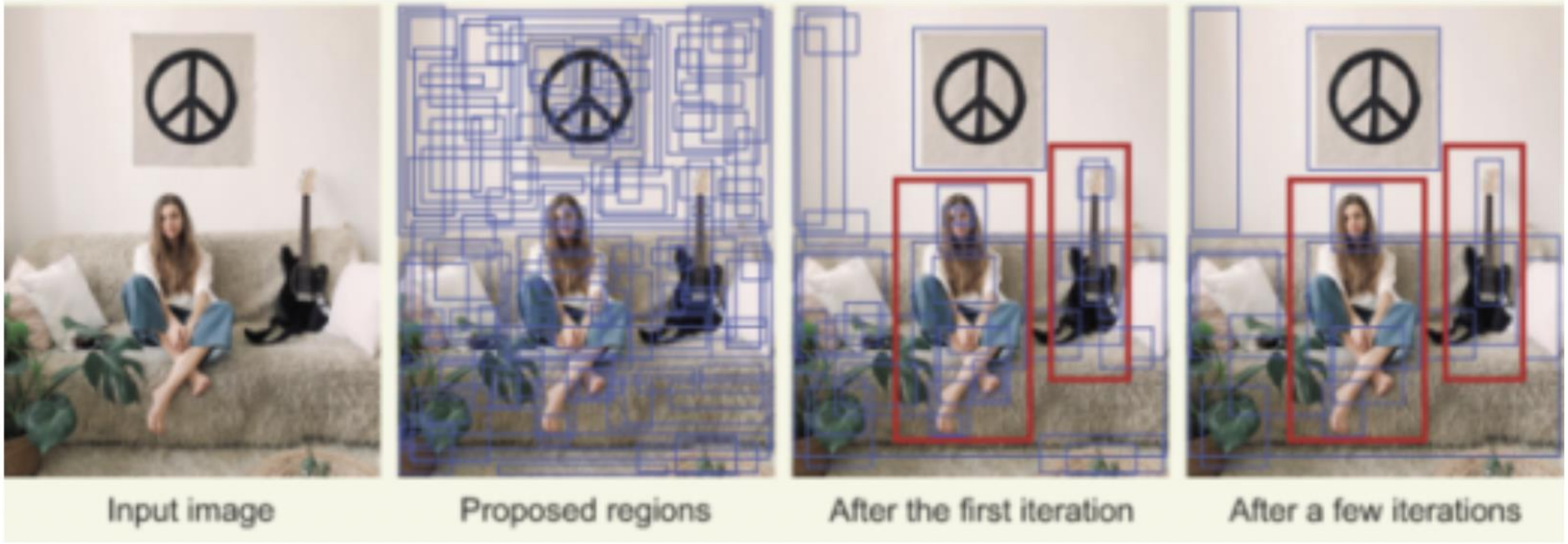

Airplane? no.
Person? yes.
TV monitor? no.
CNN
Input image
Extract regions of interest (ROI) using selective search algorithm
Warped region
A pretrained CNN to extract features
A classifier and bounding-box regressor

Selective Search Algorithm

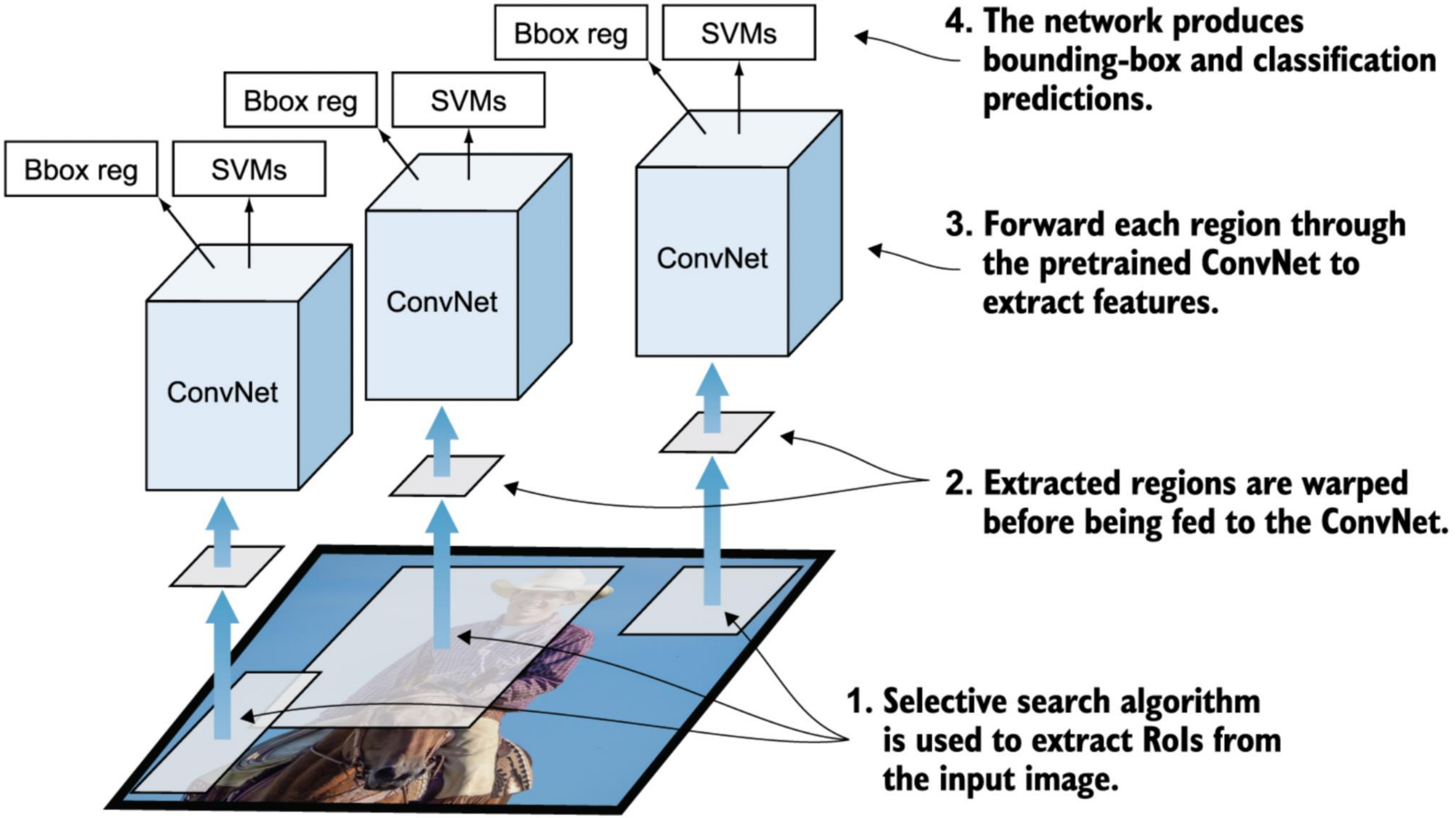
- Greedy search algorithm
- Finds an object in the images by combining similar pixels and textures into rectangular boxes
- Applies segmentation algorithm to find blobs in the image



Bottom-up segmentation using the selective search algorithm



R-CNN Architecture



Training R-CNNs

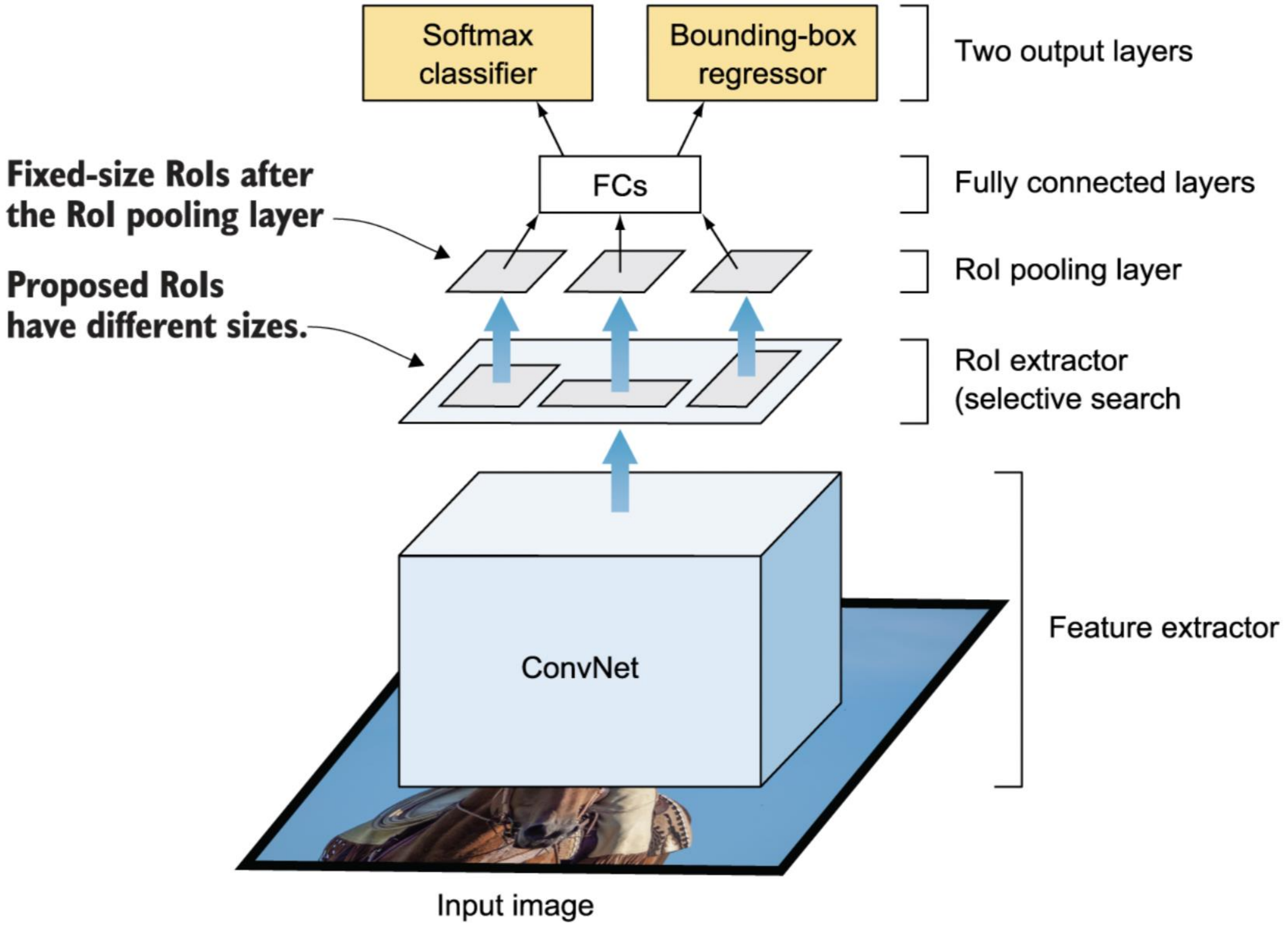
- Modules in R-CNNs
  - Region proposal
  - Feature extractor
  - Classifier
  - Bounding-box regressor
- Trainable modules
  - Feature extractor - CNN
  - Classifier – SVM
  - Bounding-box regressor
- Slow training of three separate modules without any overlap

Disadvantages of R-CNNs

- Not a single end to end system
- Performs feature extraction for each of the proposed region – not suitable for real time applications
- Multi-stage training pipeline
  - Expensive in terms of space and time

Fast R-CNN

- Improved detection speed and accuracy
- Applies CNN feature extracted on the input image before the region proposal algorithm
- Uses softmax layer instead of SVM for classification
- Uses selective search algorithm for region proposal
- RoI pooling layer – extracts a fixed-size window from the feature map



Multi-task loss function

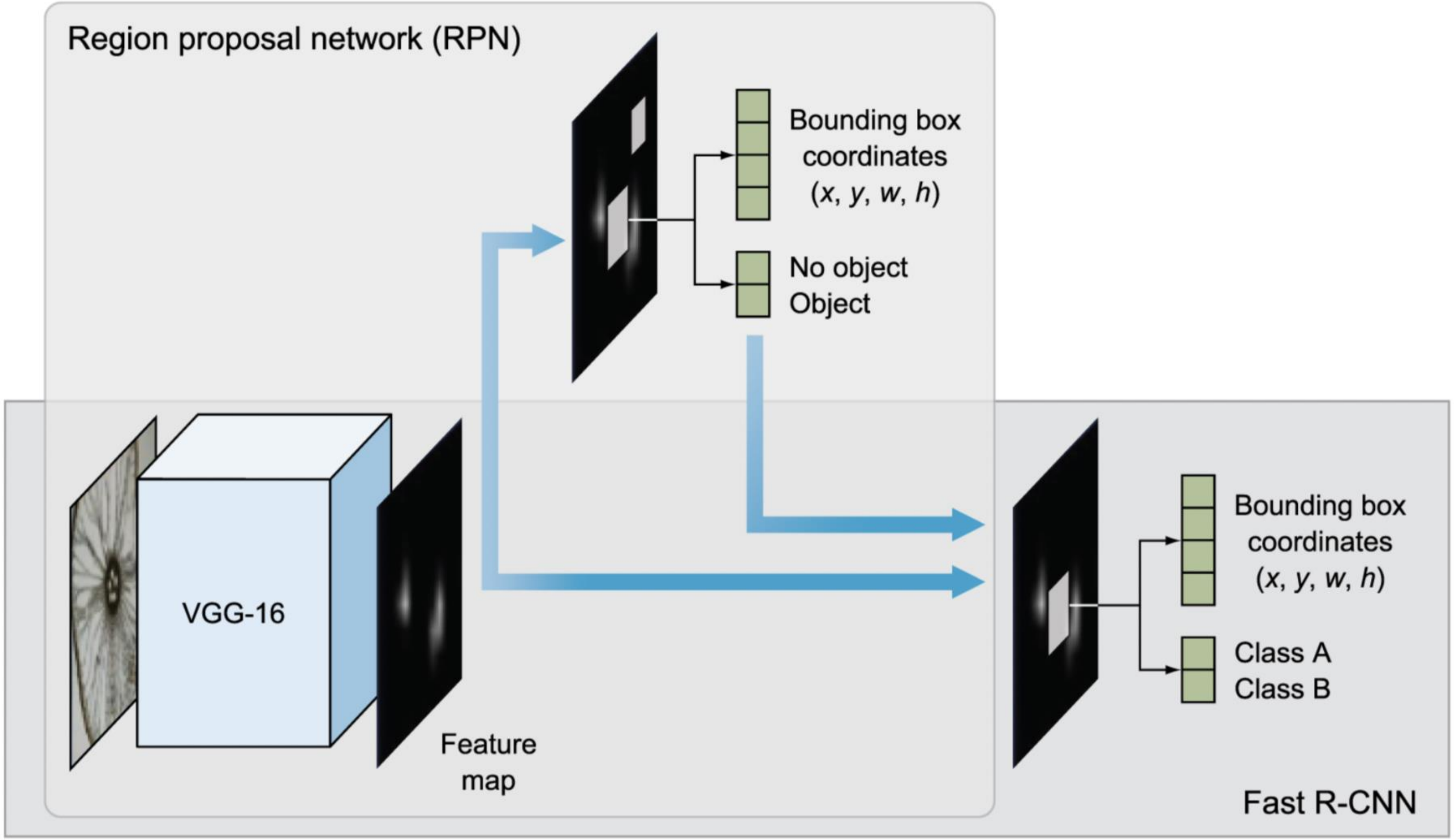
- Overall Loss function = Classification loss + Regression loss
- Classification loss, $L_{cls}(p, u) = -logp_u$ -> cross-entropy loss
- Regression loss, $L_{loc}(t^u, u) = \sum L1_{smooth}(t_i^u - v_i)$ -> absolute error

$$L(p, u, t^u, v) = L_{cls}(p, u) + [u \geq 1]\, l_{box}(t^u, v)$$

Disadvantages of Fast R-CNN

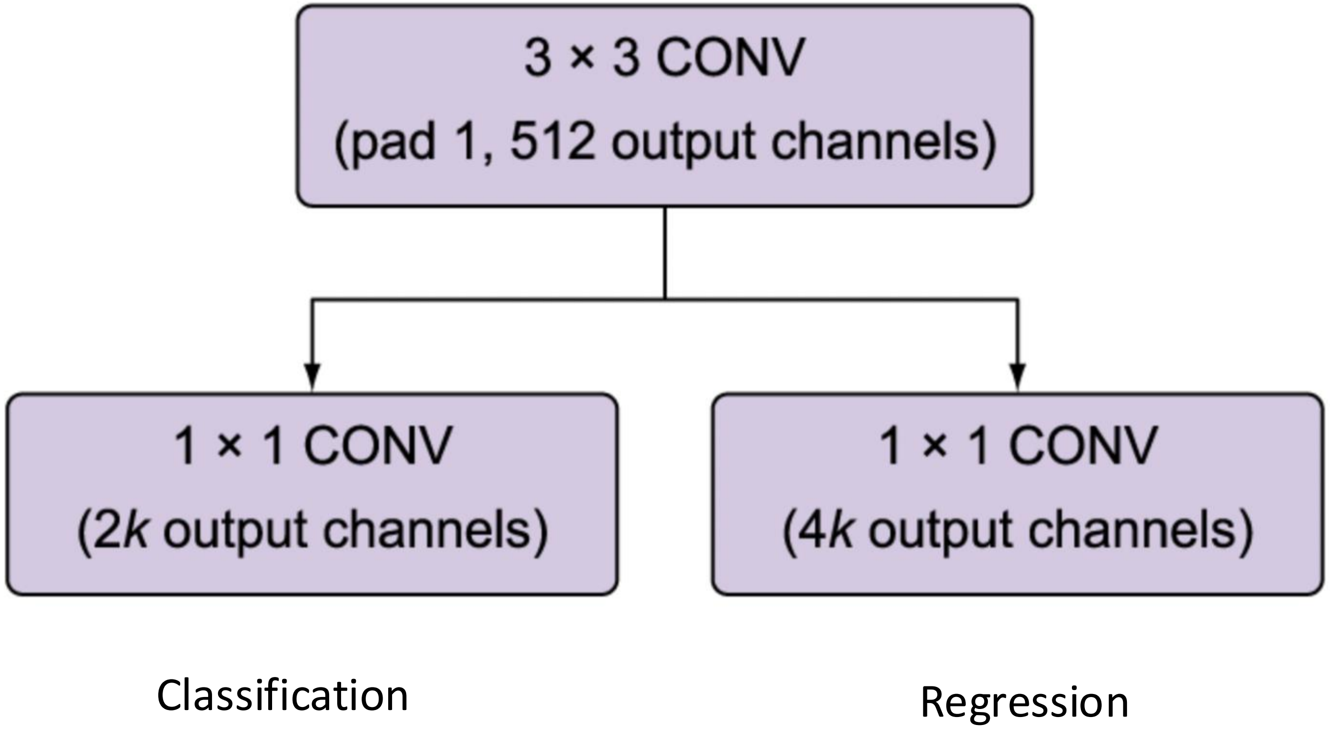
- Slower region proposal algorithm (selective search)
- Regions are proposed by a separate algorithm which is not a part of the object detection CNN

Faster R-CNN


Region proposal network (RPN)
Bounding box
coordinates
(x, y, w, h)
No object
Object
VGG-16
Feature
map
Bounding box
coordinates
(x, y, w, h)
Class A
Class B
Fast R-CNN

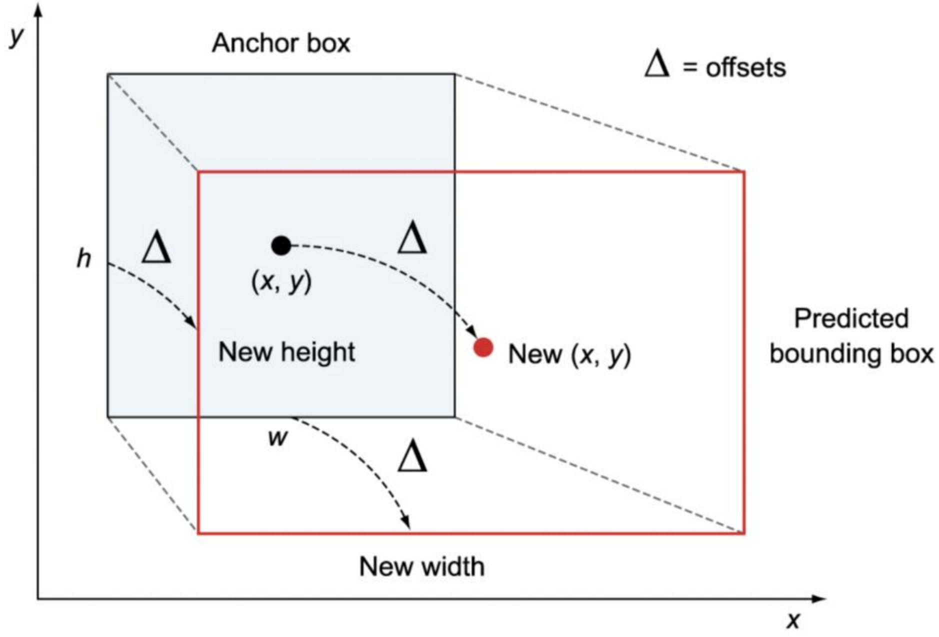
Region Proposal Network (RPN)

- Also known as attention network
- RPN classifier predicts the objectness score
- RPN regressor uses anchor boxes to predict the bounding box co-ordinates of the proposed regions

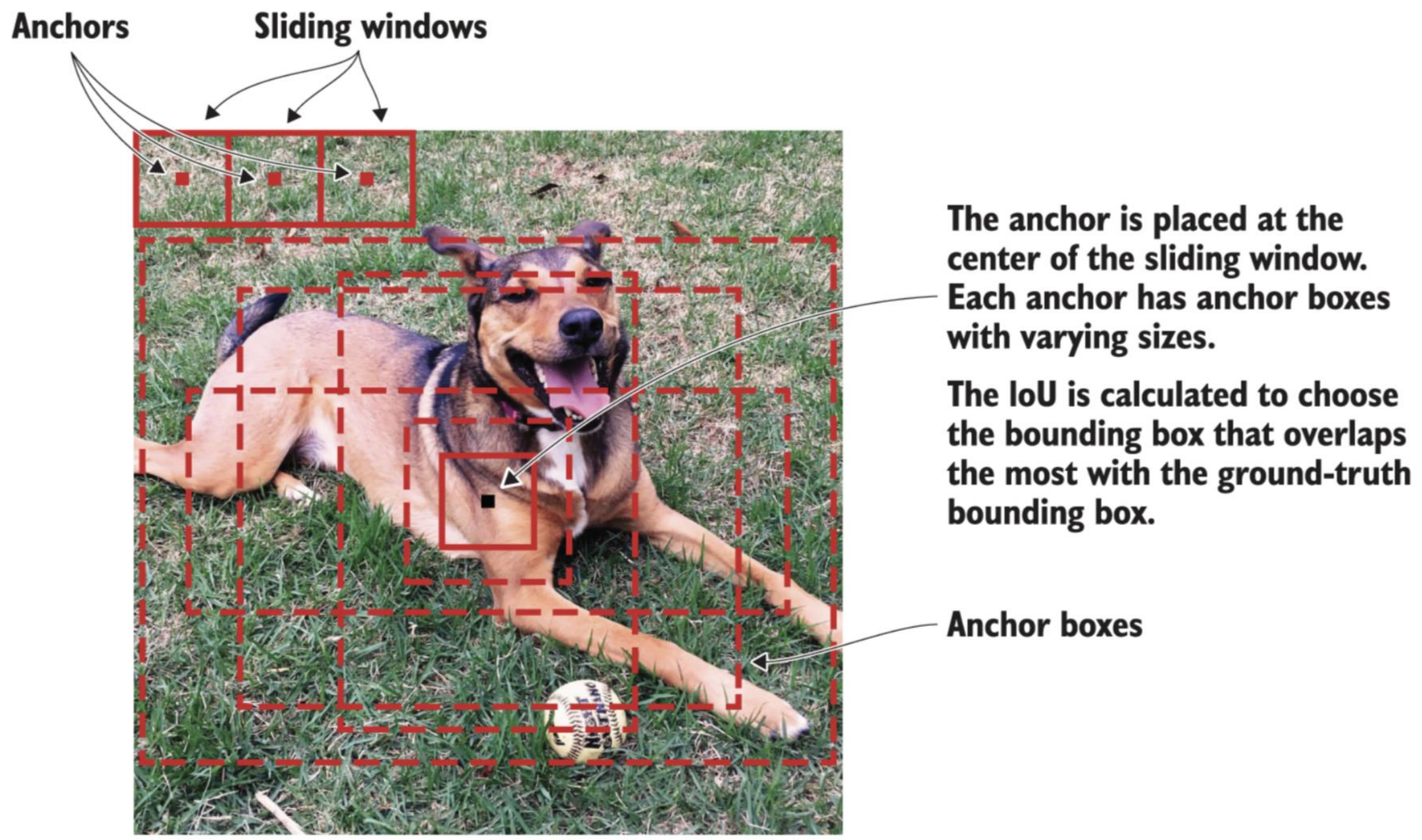


Anchor Boxes

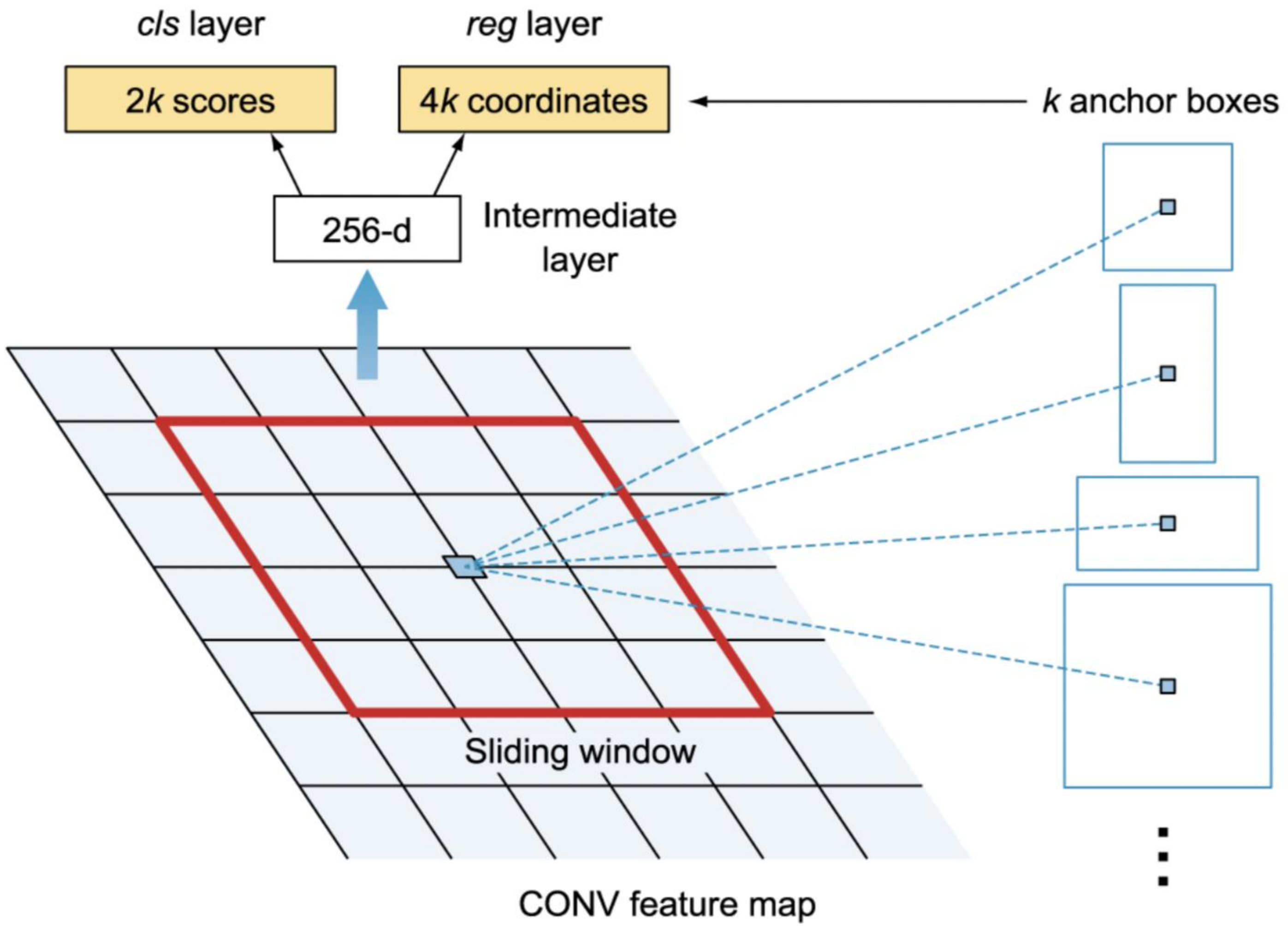
- Challenging to enforce rules to predict the center points inside the image
- Solution: create anchor boxes and predict the offsets from these boxes



Anchor Boxes



Region Proposal Network (RPN)



Limitations of Faster R-CNN

- Longer training time
- Training in multiple phases
- Slow at inference time

Reference

- Book – Deep Learning for Vision Systems by Mohamad Elgendy, Manning Publication, November 2020